

Lecture 32: Testing of Hypotheses and Interval Estimation of parameters from a SLR model.

For joint estimation of the parameters, by the least squares estimation method, we required no assumptions on the random errors. The only requirement is $\sum_{i=1}^n (x_i - \bar{x})^2 \neq 0$.

To establish unbiasedness of the least squares estimators, we need $E(\hat{\epsilon}_i) = 0 \quad \forall i = 1(n)$.

And to obtain the expressions of $\text{Var}(\hat{\beta}_0)$, $\text{Var}(\hat{\beta}_1)$ & $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$ we use, $\text{Cov}(\hat{\epsilon}_i, \hat{\epsilon}_{i'}) = \begin{cases} \sigma^2 & \text{if } i = i' \\ 0 & \text{if } i \neq i' \end{cases}$

However, for testing of hypotheses & interval estimation, we require/use $\hat{\epsilon}_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$.

Note, $\because Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $\therefore Y_i$'s are independent, but non-identical (in iid) $N(\beta_0 + \beta_1 x_i, \sigma^2)$ variables.

Then, the likelihood for $L(\beta_0, \beta_1, \sigma^2 | y, z) = (2\pi\sigma^2)^{-n/2} \exp. -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$

For fixed σ^2 , maximizing the likelihood wrt β_0 & β_1 $\Rightarrow$ For fixed σ^2 ,
 $\Leftrightarrow$ minimizing $\sum (y_i - \beta_0 - \beta_1 x_i)^2$ wrt β_0 & β_1 $\Rightarrow \hat{\beta}_0(\text{MLE}) = \hat{\beta}_0(\text{LSE})$
 and $\hat{\beta}_1(\text{MLE}) = \hat{\beta}_1(\text{LSE})$

To obtain mle of the remaining parameter σ^2 , consider

the profile likelihood $\hat{L}(\hat{\beta}_0, \hat{\beta}_1, \sigma^2 | y, z)$ or $\ln L(\hat{\beta}_0, \hat{\beta}_1, \sigma^2 | y, z)$ & maximize it wrt σ^2

$$\Rightarrow \frac{\partial \ln L}{\partial \sigma^2} \Big|_{\hat{\sigma}^2_{\text{MLE}}} = 0 \Rightarrow \frac{\partial}{\partial \sigma^2} \left[\frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2\sigma^2} \text{SSR}_{\text{OLS}} \right] \Big|_{\hat{\sigma}^2_{\text{MLE}}} = 0$$

$$\Rightarrow \hat{\sigma}^2_{\text{MLE}} = \frac{\text{SSR}_{\text{OLS}}}{n} \approx \hat{\sigma}^2_{\text{UE}} = \frac{\text{SSR}_{\text{OLS}}}{n-2}$$

In all testing problems, we take $\hat{\sigma}^2 = \hat{\sigma}^2_{\text{UE}}$

From $Y_i \stackrel{\text{iid}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$, it follows that

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2}\right)\right), \hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum x_i^2}\right), \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

Further $\hat{\beta}_0$ & $\hat{\sigma}^2$ are independently distributed. So are $\hat{\beta}_1$ & $\hat{\sigma}^2$.

(Use the least sq. estm form of $\hat{\beta}_0$ & $\hat{\beta}_1$ to express them as linear fun of the responses $y_1, \dots, y_n$.)

2) Use the Jacobian/mgf method to prove normality.

3) Exp., variance & cov. etc. are already established)

I To test

$$H_0: \beta_0 = \beta_0^* \text{ (fixed)} \text{ against } H_A: \beta_0 \neq \beta_0^*.$$

Use, $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}))$, $(n-2)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$ indep, so that

$$t = \frac{\hat{\beta}_0 - \beta_0^*}{\left[\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right]^{1/2}} \sim t_{n-2} \text{ under } H_0. \quad \left(\text{This is important since we're replacing } \beta_0 \text{ by } \beta_0^* \text{ as given in } H_0 \right)$$

The denominator of the above t-statistic is called the Standard Error of $\hat{\beta}_0$ (s.e.) (= sq. root of estimated $\text{Var}(\hat{\beta}_0)$) as against std. deviation ($\hat{\beta}_0$), which is the sq. root of the variance ($\hat{\beta}_0$).

The critical regions for the diff. H_A are

$$w: \text{obsd } t > t_{n-2, \alpha}$$

$$< t_{n-2, \alpha} (= -t_{n-2, 1-\alpha})$$

$$|t| > t_{n-2, \alpha/2} \quad (\text{or, obsd } t > t_{n-2, \alpha/2} \text{ or, obsd } t < t_{n-2, 1-\alpha/2})$$

Note, that the t-statistic defined above, can be used as a pivot for obtaining the $100(1-\alpha)\%$ confidence interval of β_0 , in the following manner:

$$P \left[\frac{|\hat{\beta}_0 - \beta_0|}{\text{s.e.}(\hat{\beta}_0)} \leq t_{n-2, \alpha/2} \right] = 1 - \alpha$$

$$\Rightarrow \text{A } 100(1-\alpha)\% \text{ C.I. for } \beta_0 \text{ is } \hat{\beta}_0 \pm t_{n-2, \alpha/2} \cdot \text{s.e.}(\hat{\beta}_0)$$

$$\text{Using, } \hat{\beta}_0 \sim N(\beta_0, \text{Var}(\hat{\beta}_0)) \quad (n-2)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2} \text{ indep.}$$

II. To test $H_0: \beta_1 = \beta_1^*$ ag. $H_A: \beta_1 \neq \beta_1^*$

Use, $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$, $(n-2)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$ indep, so that

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\text{s.e.}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0 \quad \left(\text{s.e.}(\hat{\beta}_1) = \left[\hat{\sigma}^2 \frac{1}{S_{xx}} \right]^{1/2} \right)$$

$$w: \text{obsd } t > t_{n-2, \alpha} < t_{n-2, 1-\alpha} (= -t_{n-2, \alpha})$$

$$|t| > t_{n-2, \alpha/2}$$

$$\text{A } 100(1-\alpha)\% \text{ C.I. for } \beta_1 \text{ is given by } \hat{\beta}_1 \pm t_{n-2, \alpha/2} \cdot \text{s.e.}(\hat{\beta}_1)$$

Next, Bayesian inference.